

PROJECT DOCUMENTATION

Team Information

Team Name: ScamShield

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Problem Statement

Explainable Detection of Scam and Phishing Messages

Scam and phishing messages often use urgency, threats, financial requests, rewards, and verification demands to manipulate users into taking unsafe actions. Users may find it difficult to identify these warning signs before clicking links, sharing information, or responding.

ScamShield addresses this problem by analyzing suspicious message signals and providing an explainable risk assessment rather than simply returning a safe/unsafe verdict.

Event

HackMatrix 2026 — Round 2

Links

GitHub Repository:

<https://github.com/AI-Engineer000/X-HackMatrix2026>

Live Deployed Link:

Not publicly deployed. Prototype demonstrated through local Flask deployment.

Demo Video:

Project Details

Project Name / Title

ScamShield — Explainable Message Risk Analysis

Platform

Web Application

Preview

ScamShield provides a web interface where users can submit suspicious SMS, email, or direct-message content. The system analyzes the message and displays its risk level, **0–100** risk score, detected warning signals, supporting evidence, and a safety recommendation.

The application also maintains a recent scan history using SQLite.

Summary

ScamShield is a lightweight, explainable cybersecurity prototype designed to help users understand potentially fraudulent messages before interacting with them.

Rather than presenting only a binary verdict, ScamShield identifies suspicious signals and presents the reasoning behind the assessment in a user-friendly format.

Problem Being Solved

Users frequently encounter scam and phishing messages that create urgency, fear, or financial pressure. Identifying such messages manually can be difficult, especially when the message appears legitimate.

ScamShield helps users recognize suspicious characteristics and make more informed decisions before responding, clicking links, or sharing sensitive information.

USP — Unique Selling Point

Explainability-first scam detection.

ScamShield does not only provide a risk level. It shows:

Risk Score → Detected Signal → Evidence → Recommendation

This allows users to understand *why* a message has been classified as risky.

Key Features

- Message-based scam risk analysis
 - LOW / MEDIUM / HIGH risk classification
 - 0–100 explainable risk score
 - Detection of suspicious message signals
 - Evidence associated with detected signals
 - Safety recommendations
 - Recent scan history
 - Local SQLite storage
 - Input validation and error handling
 - Simple and responsive web interface
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Tech Stack

Frontend: HTML5, CSS3, JavaScript

Backend: Python, Flask

Database: SQLite

Analysis: Signal / Rule-Based Message Analysis

Data Handling: JSON

Development: Git & GitHub

Future Scope

- Machine-learning and NLP-based scam classification
- Multilingual scam-message detection
- Real-time URL reputation checking
- Phishing-domain intelligence
- Screenshot/image-based scam detection
- Browser and messaging-platform integration
- Threat-intelligence integration
- Advanced analytics and reporting